Diagrama de dimensiones de la antena.

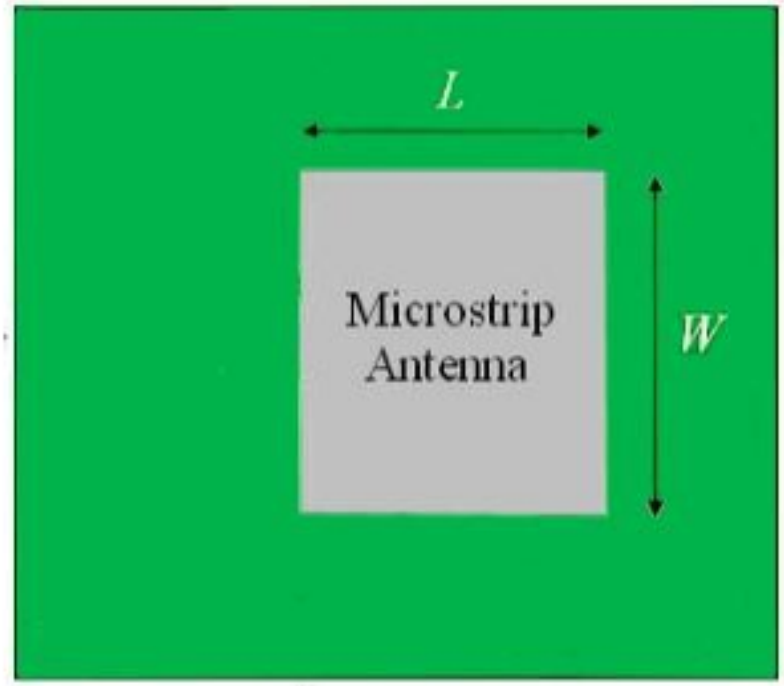

Figura 2-2-1. Geometría de una línea microstrip.

Diagrama de dimensiones de la línea.

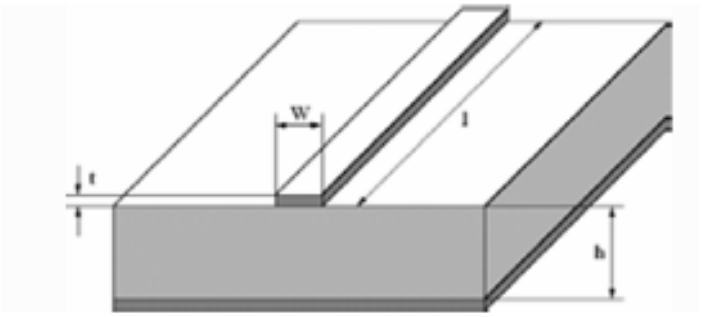

Figura 2-4-1. Geometría de una línea microstrip.

Microstrip vs Stripline.

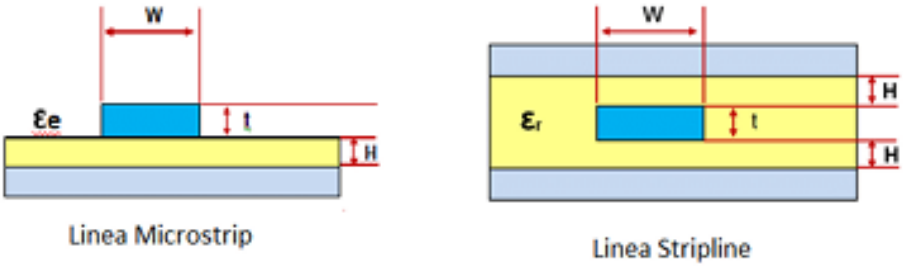

Figura 2-4-2. Línea Microstrip y Stripline.

Figura 4-1. Esquema de la antena con vista lateral.

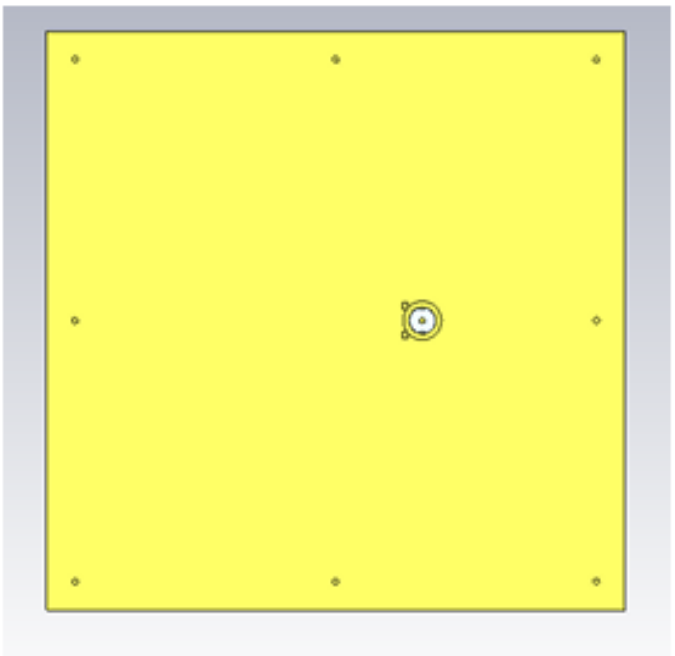

Figura 4-2. Conector y plano de masa.

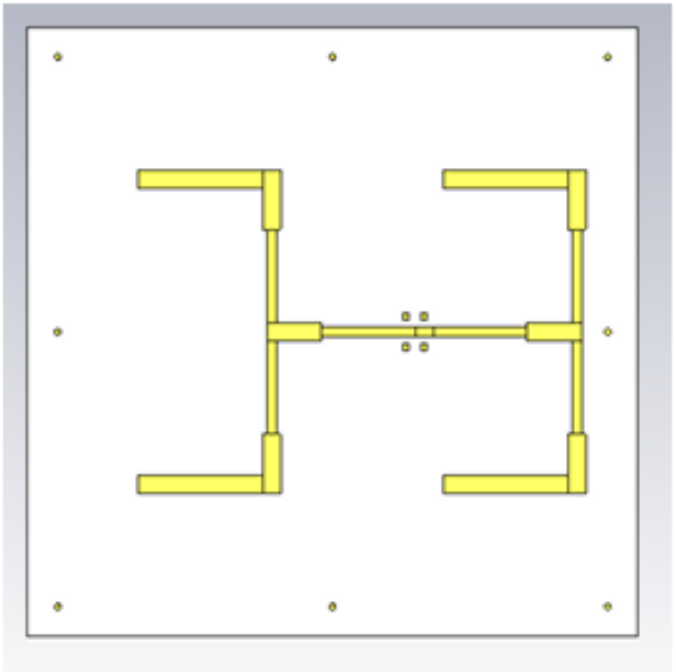

Figura 4-3. Guías de alimentación.

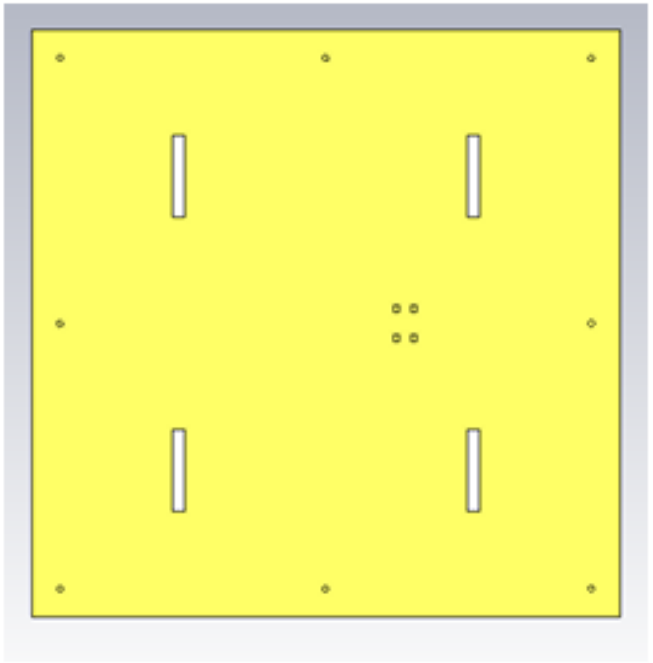

Figura 4-4. Plano de ranuras.

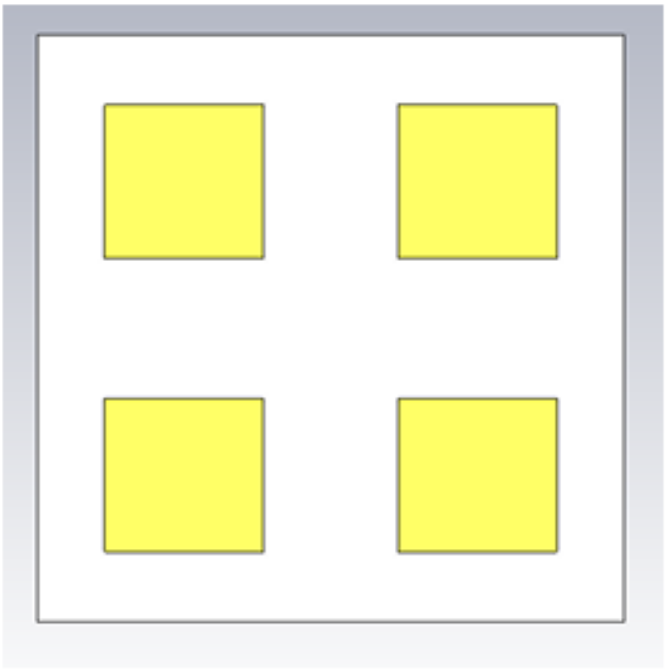

Figura 4-5. Antenas path.

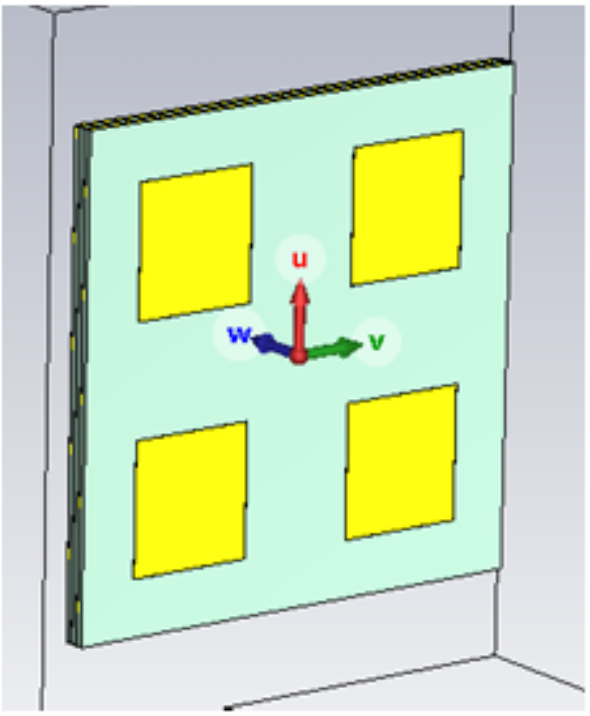

Figura 4-6. Antena patch - perfil frontal

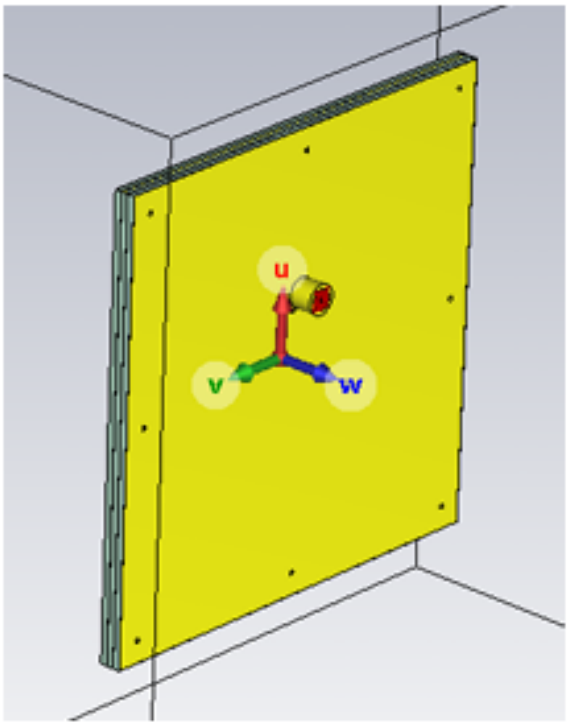

Figura 4-7. Antena patch - perfil posterior

Coeficiente de reflexión de la antena.

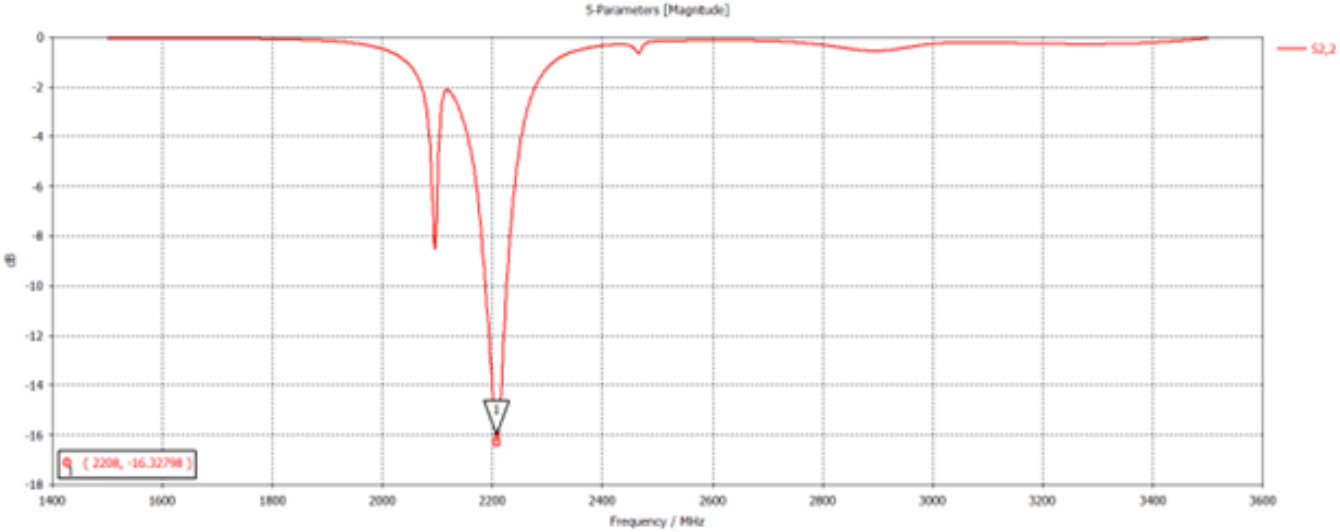

Figura 5-1. Gráfica de Parámetros S en CST.

Coeficiente de reflexión de la antena - Formato de carta de Smith.

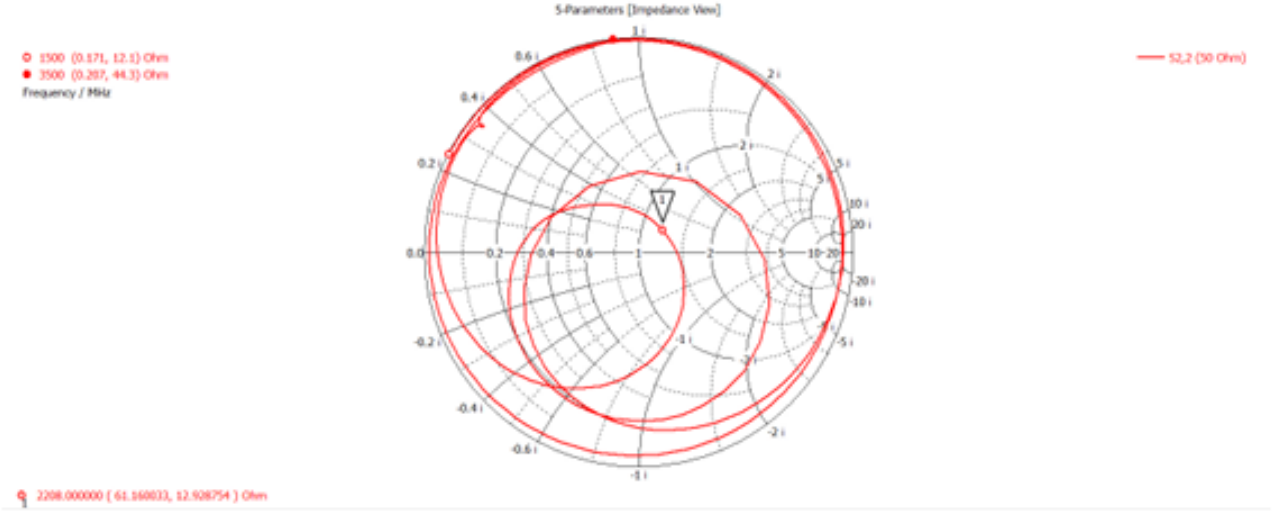

Figura 5-2. Gráfica de Carta de Smith en CST.

Relación onda estacionaria de la antena.

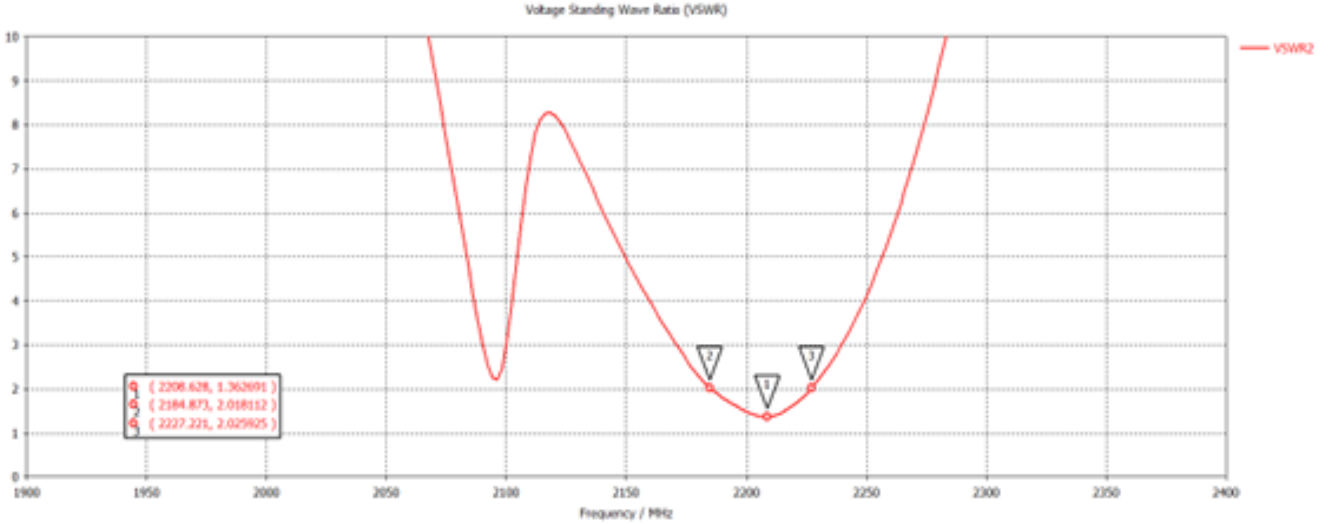

Figura 5-3. Gráfica del SWR en CST - Ancho de banda para con roe=2 : 43[MHz] (entre los puntos 2 y 3).

Vista del farfield de la antena

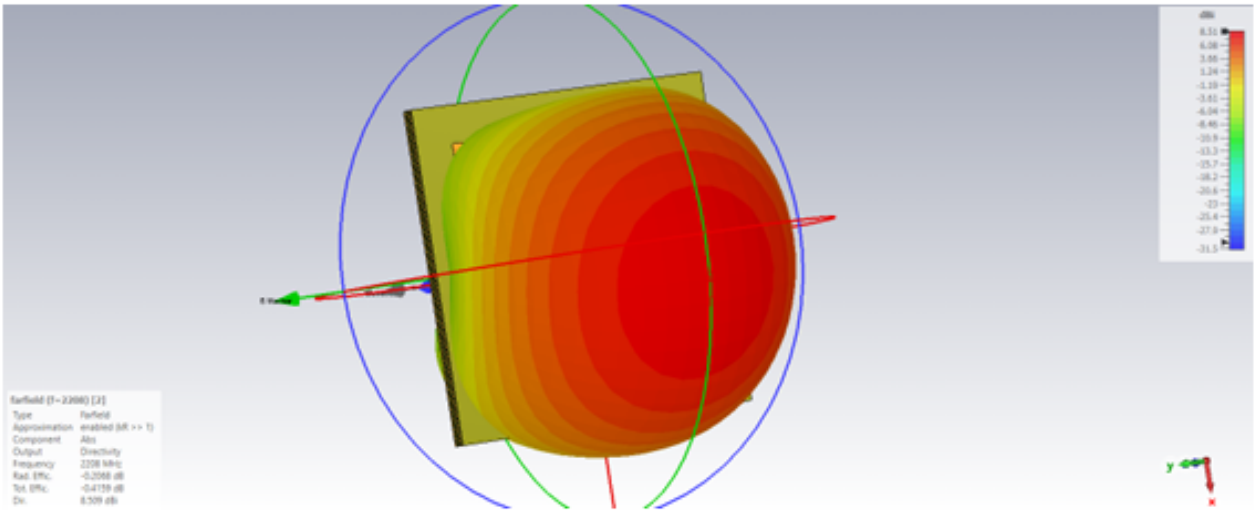

Figura 6-4. Gráfica del Fairfield para 2208 MHz en CST

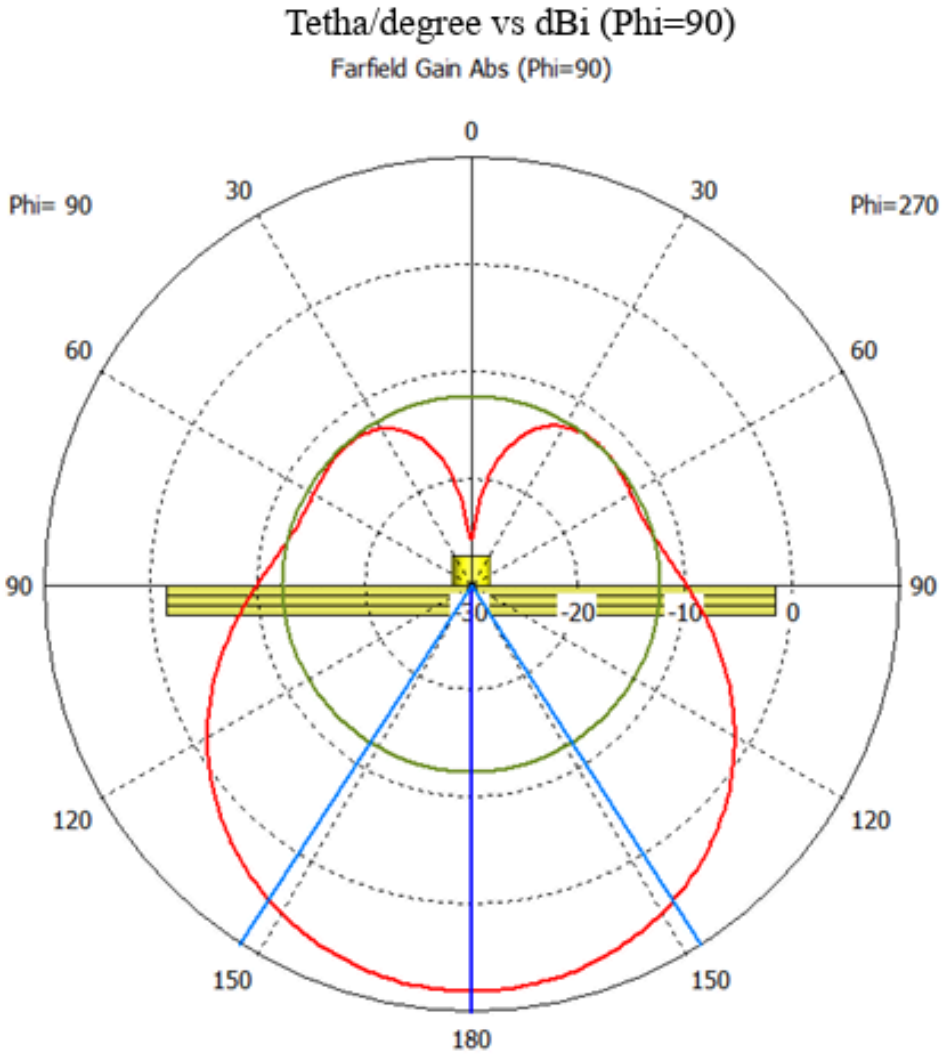

Figura 5-5. Gráfica polar de la antena [Cut angle: 90 - Constant: Phi]

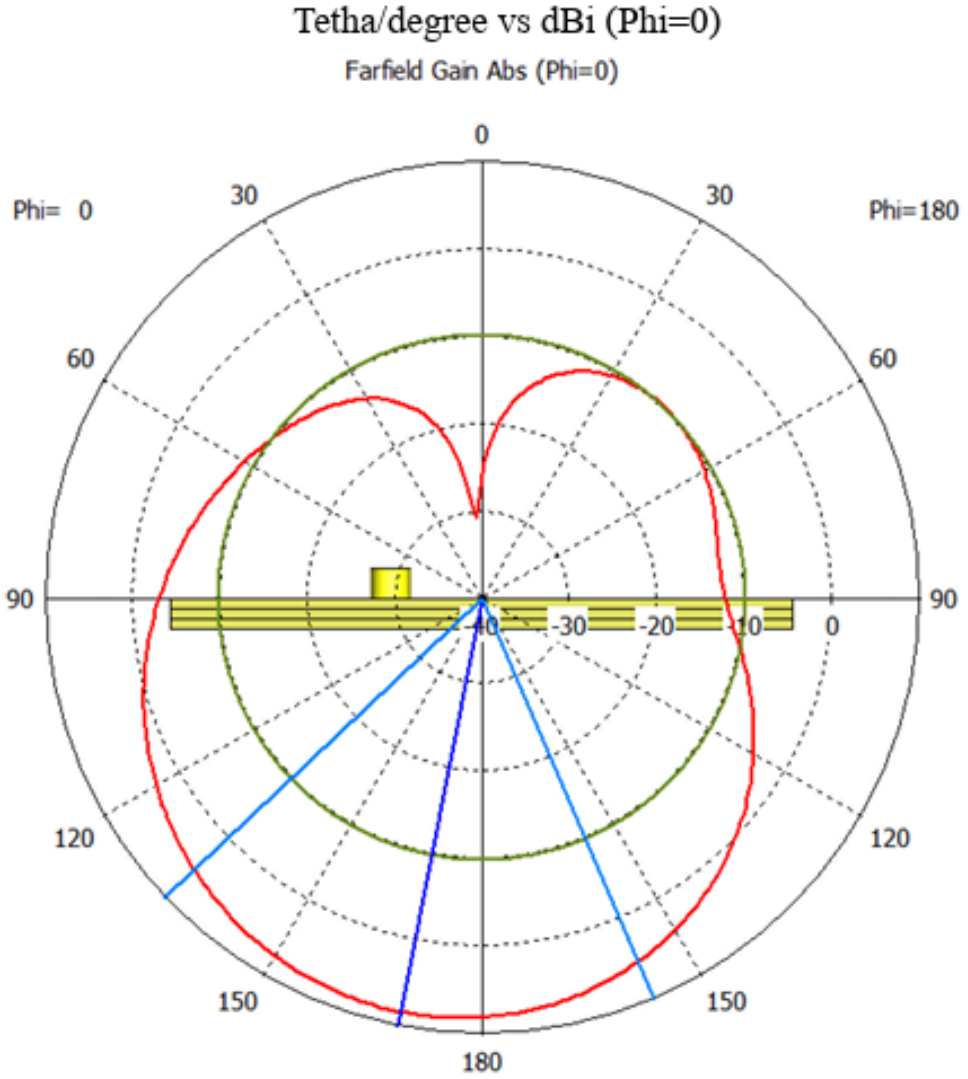

Figura 5-6. Gráfica polar de la antena [Cut angle: 0 - Constant: Phi]

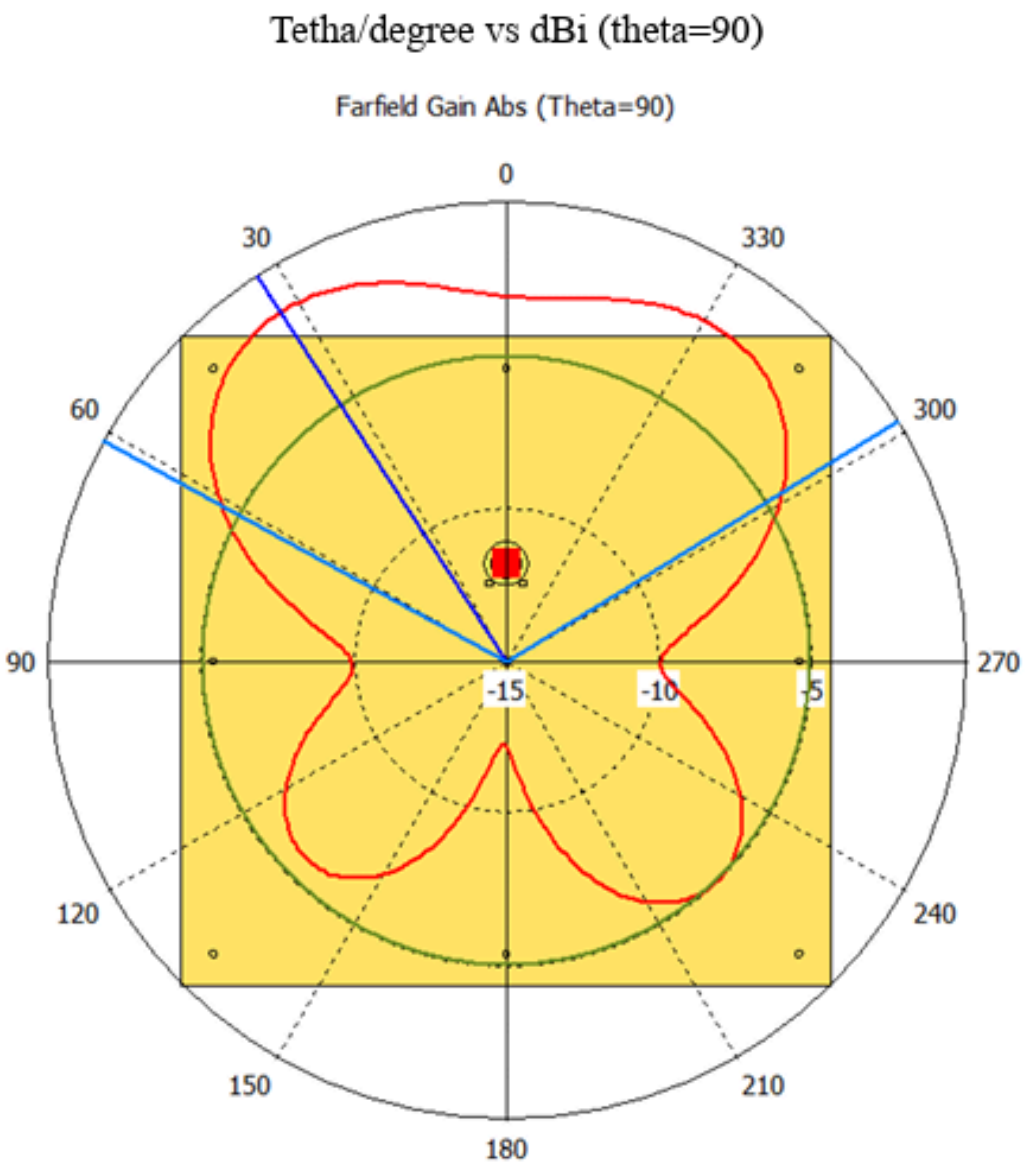

Figura 5-7. Gráfica polar de la antena [Cut angle: 90 - Constant: Theta]

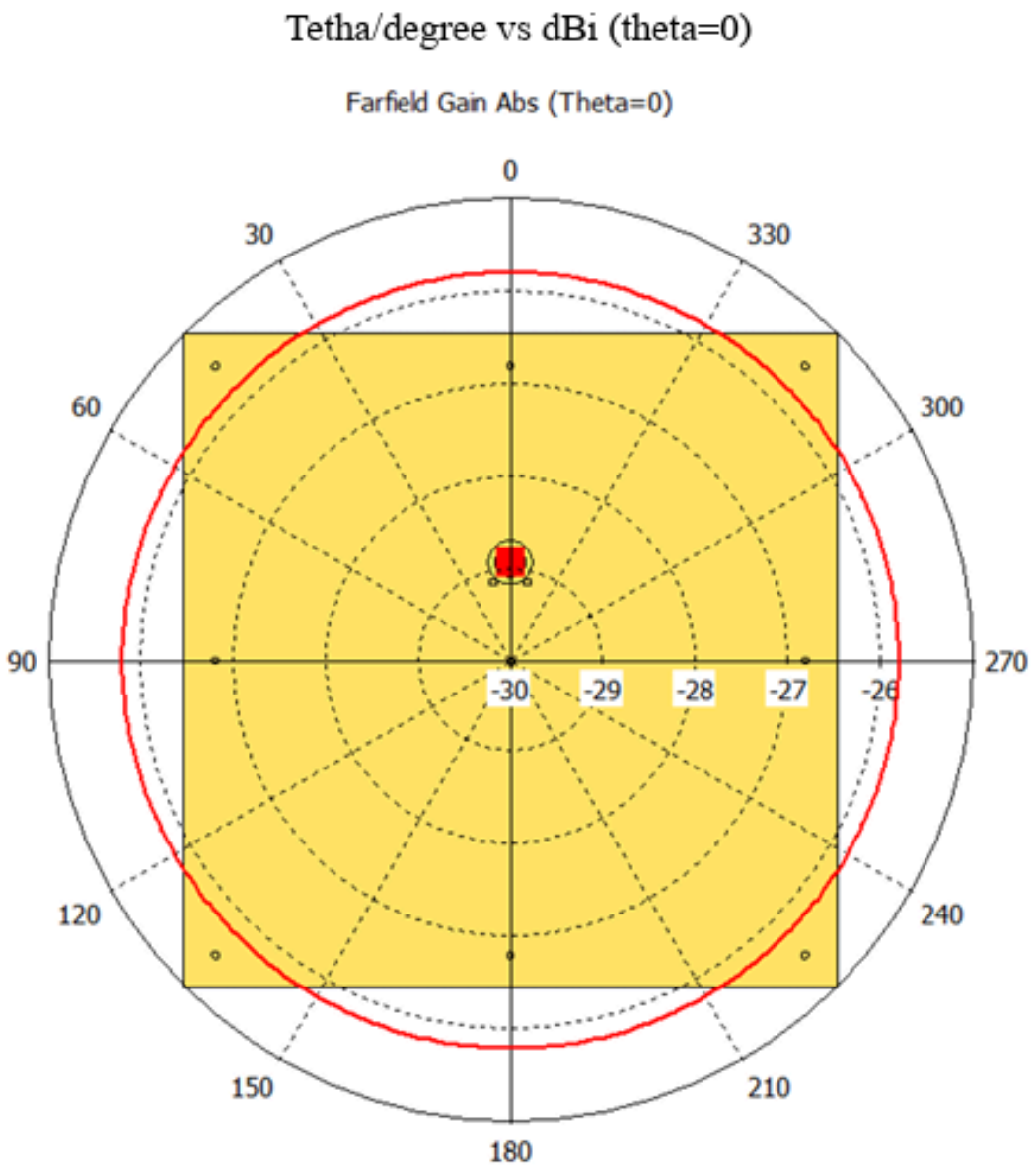

Figura 5-8. Gráfica polar de la antena [Cut angle: 0 - Constant: Theta]

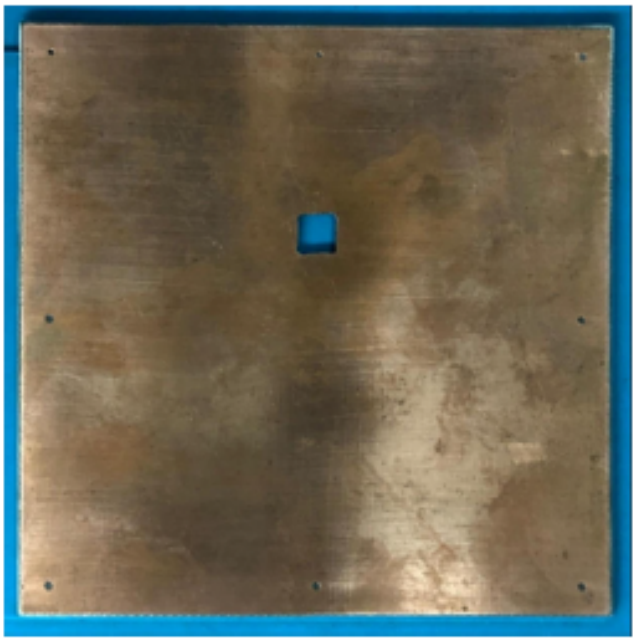

Figura 6-1. Masa de la antena.

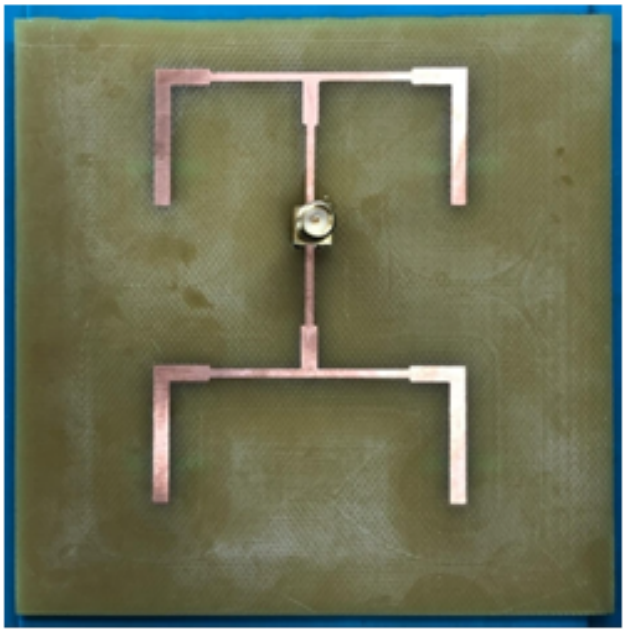

Figura 6-2. Guías de alimentación.

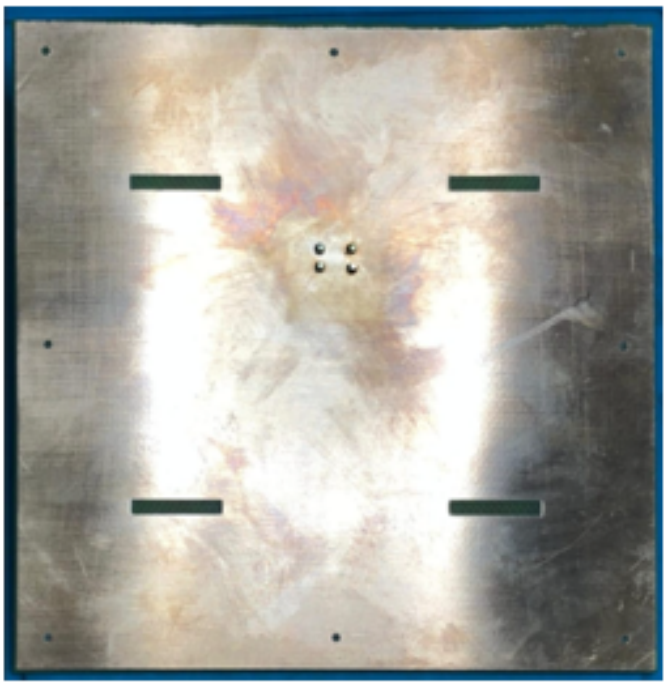

Figura 6-3. Capas de ranuras.

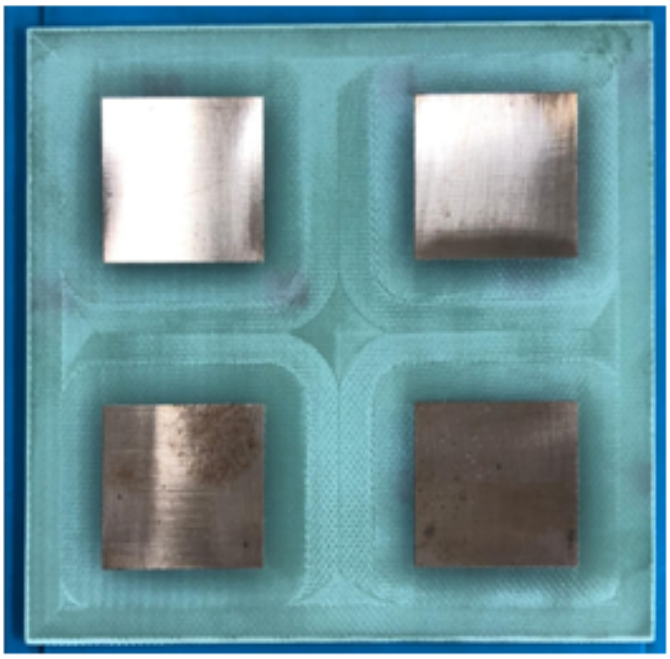

Figura 6-4. Antenas patch.

Figura 6-5. Resultado final luego de soldar los bastones de cobre.

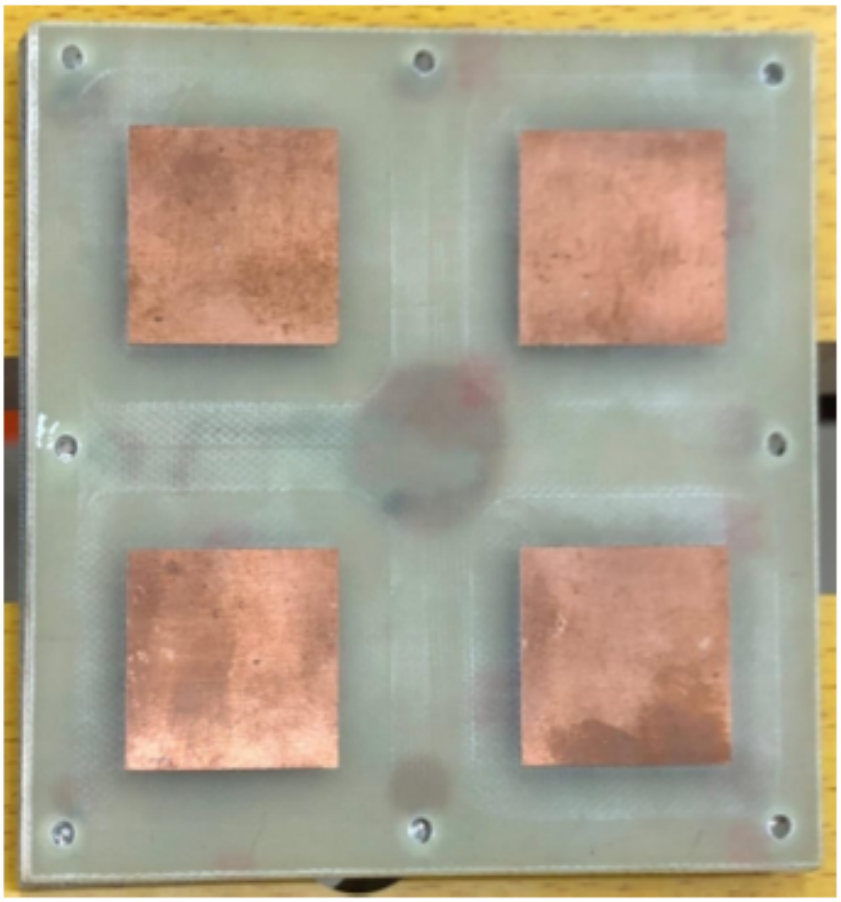

Figura 6-6. Vista frontal de la antena patch.

Coeficiente de reflexión de la antena.

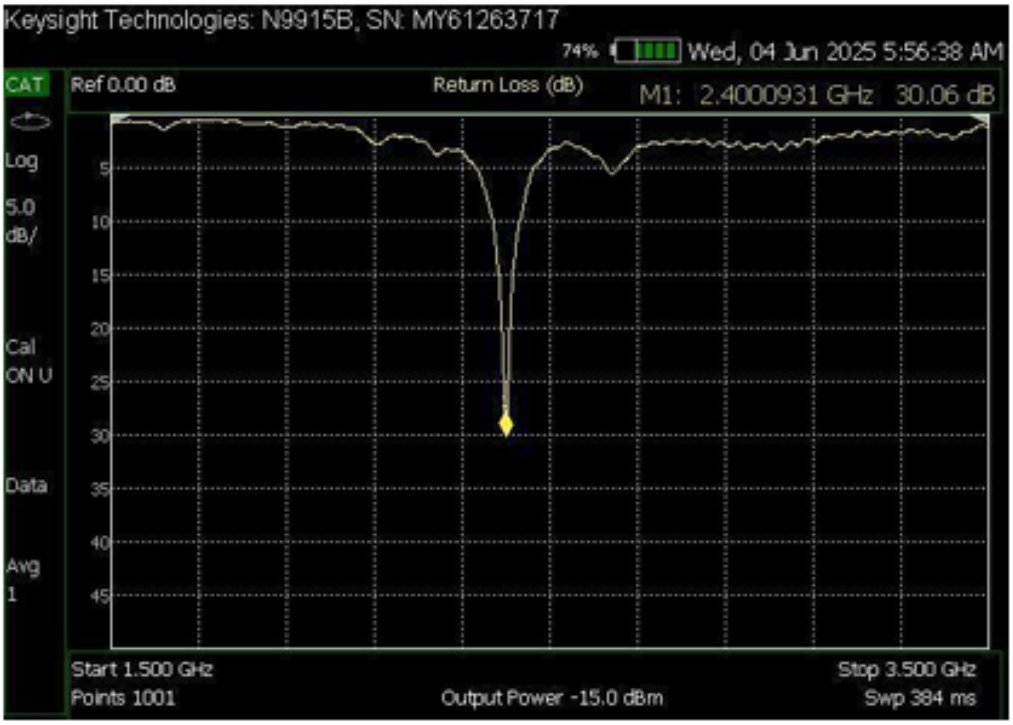

Figura 6-1-1. Pérdidas por retorno - VNA.

Relación onda estacionaria de la antena.

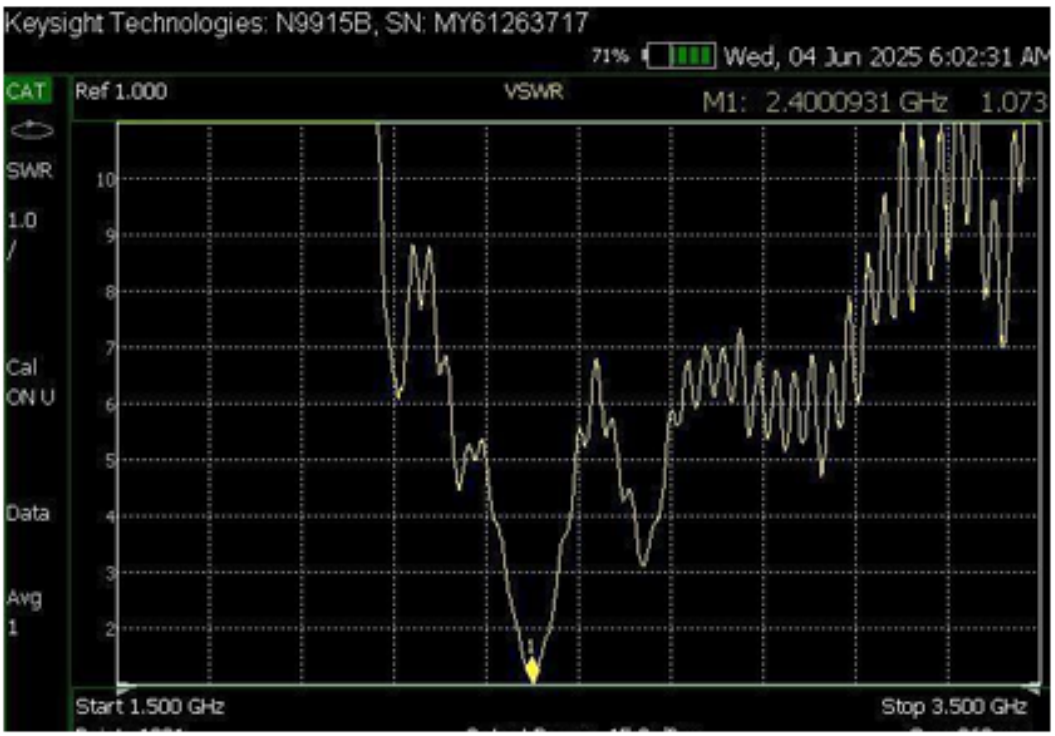

Figura 6-1-2. Relación onda estacionaria (ROE) - VNA.

Coeficiente de reflexión de la antena - Formato de carta de Smith.

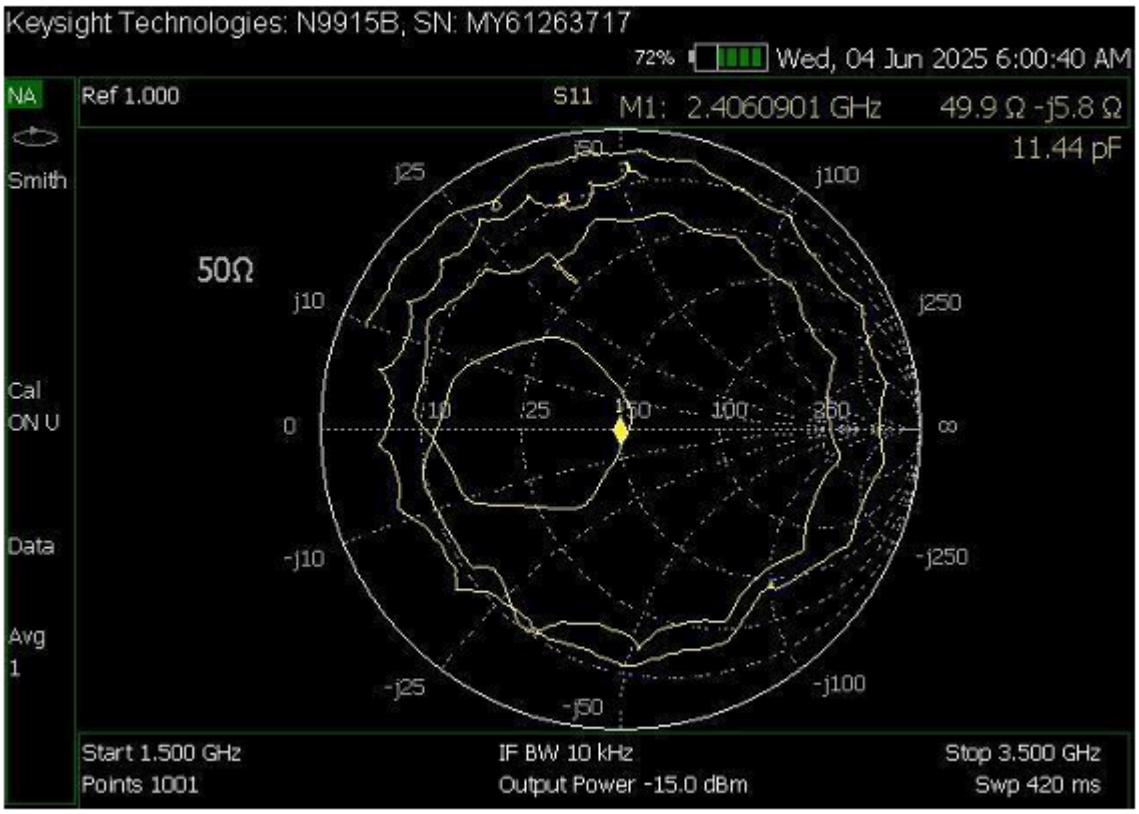

Figura 6-1-3. Carta de Smith.

Coeficiente de reflexión simulador vs frecuencia.

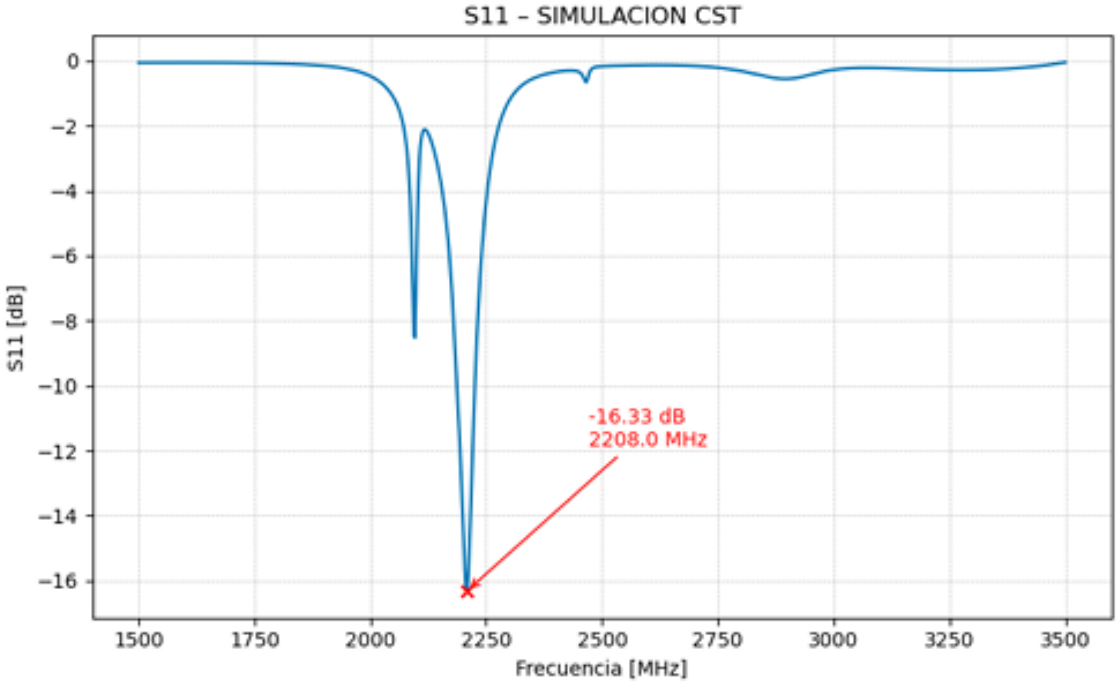

Figura 6-2-1. S11 de la antena simulada en CST

Coeficiente de reflexión VNA vs frecuencia.

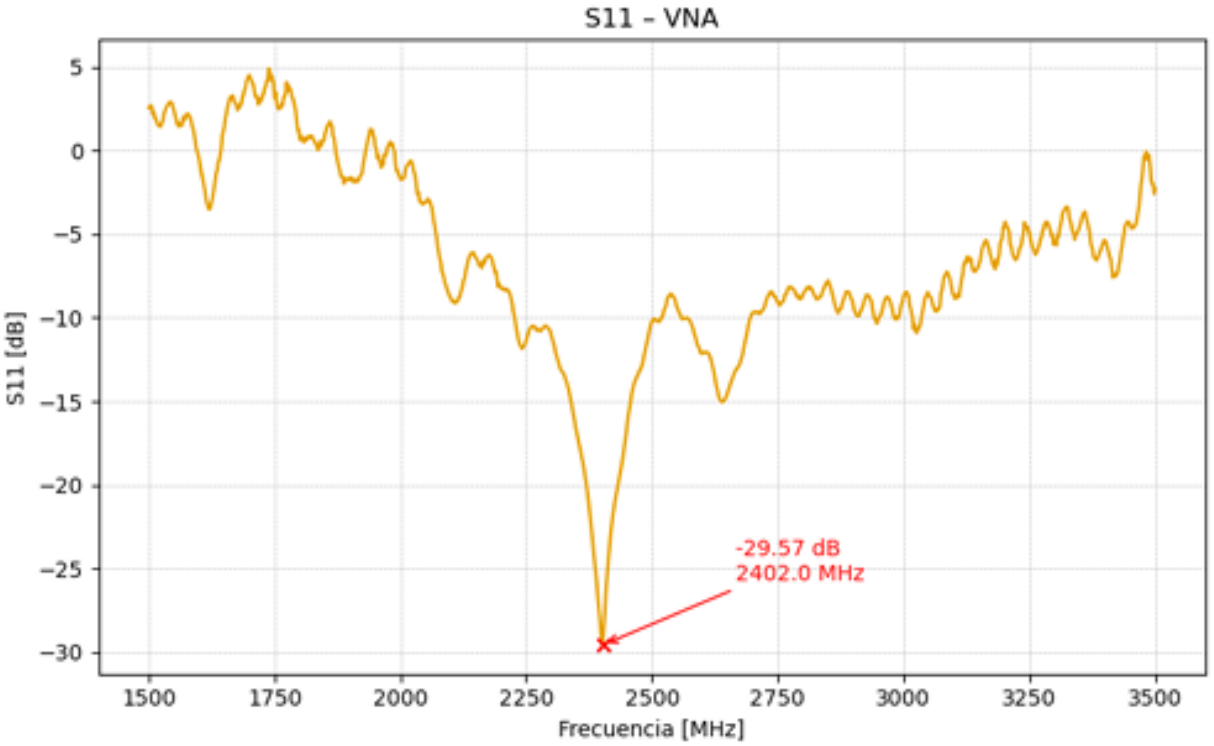

Figura 6-2-2. S11 de la antena real medida con el VNA

Coeficiente de reflexión comparado entre el simulador y el VNA.

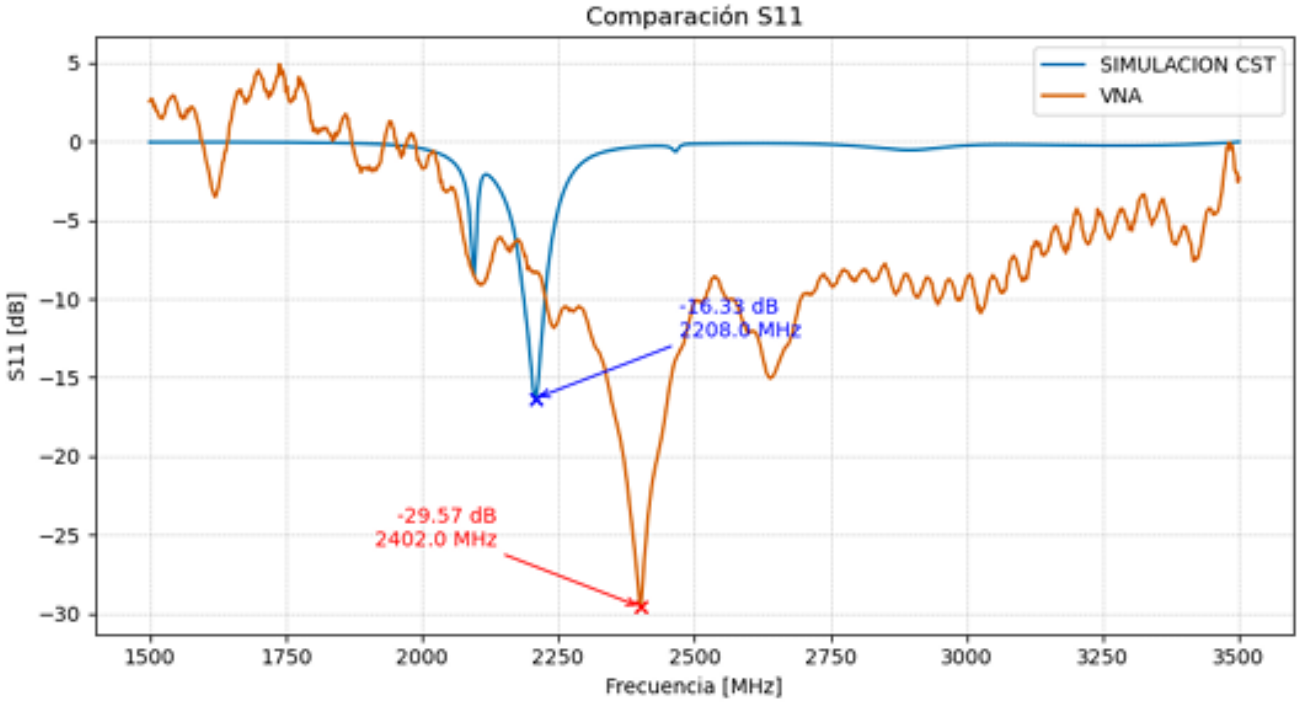

Figura 6-2-3 .Comparación de los S11

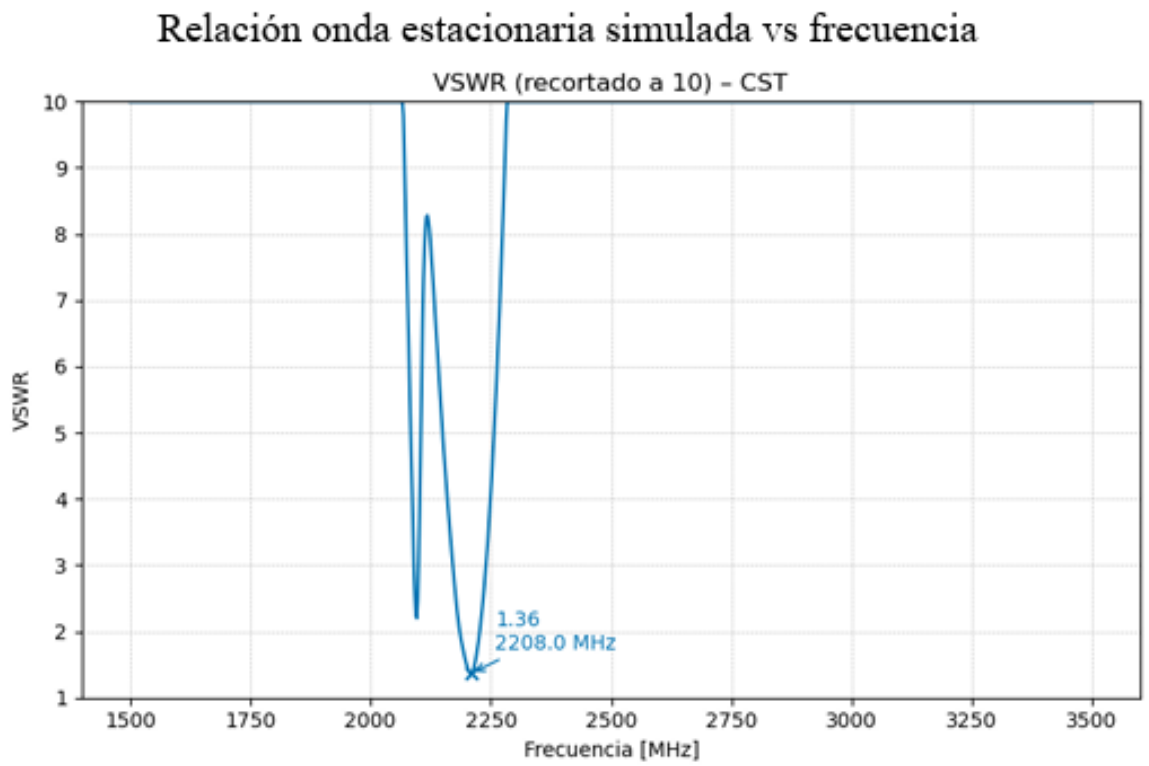

Figura 6-2-4. SWR de la antena simulada en CST

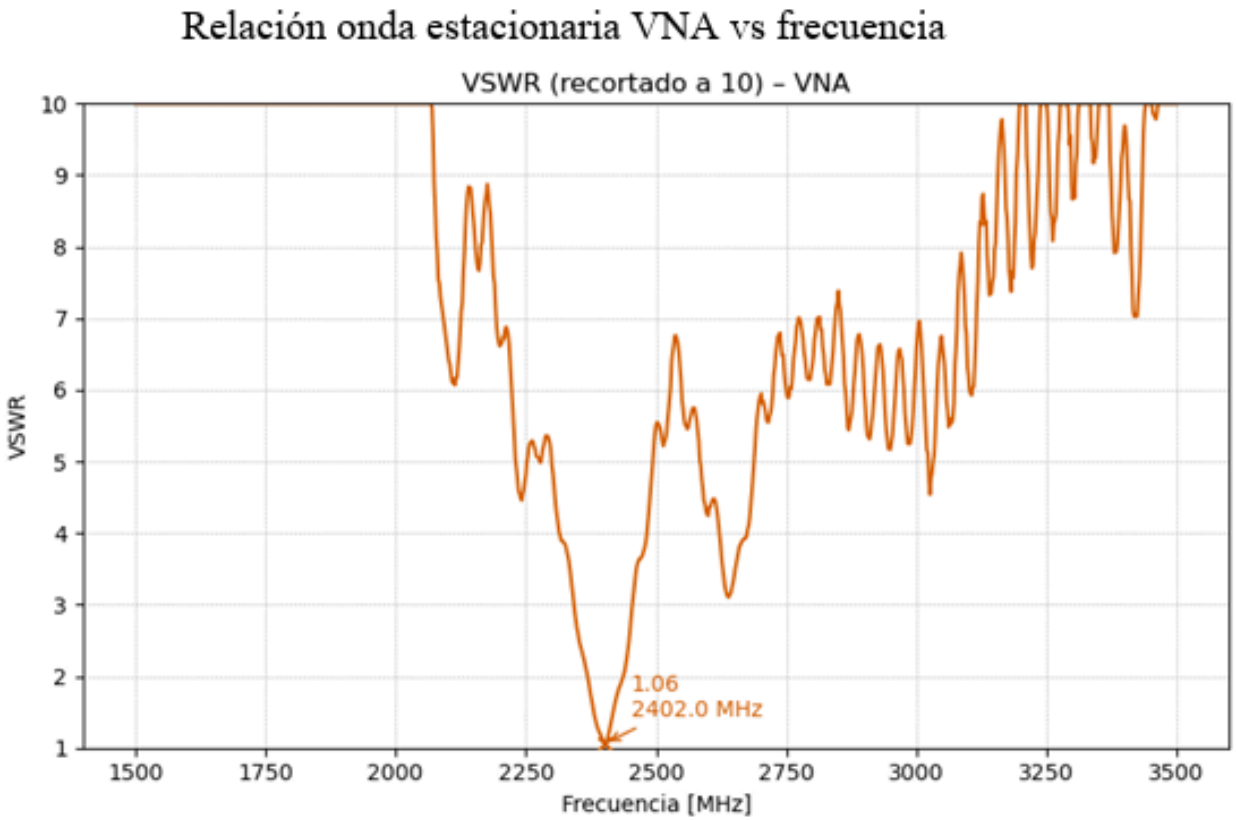

Figura 6-2-5. SWR de la antena real medida con el VNA

Comparación de la relación onda estacionaria entre el simulador y el VNA

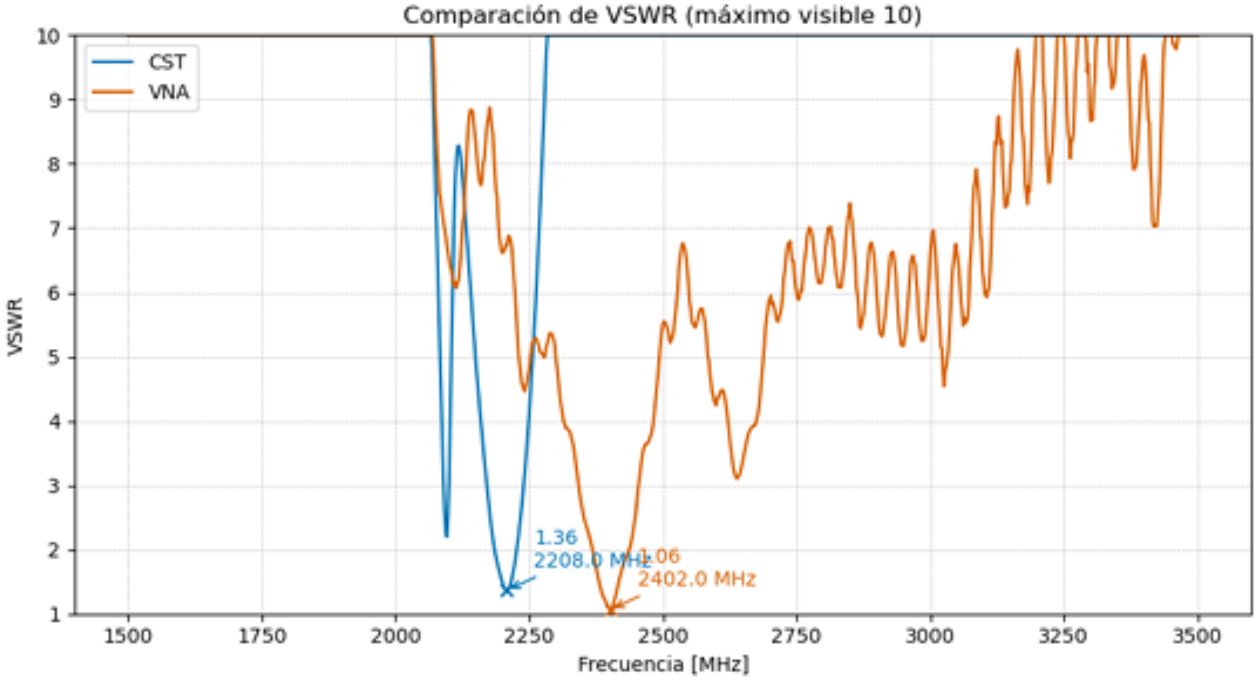

Figura 6-2-6. Comparación de los SWR